

Q1. Answers to the following questions should be brief, clear, and complete:

- What is the difference between a DBMS and an IR System?
- Define *support* and *confidence* in the context of association rule mining?
- Do we need training data for classification? If so then why? Do we need training data for clustering? If so then why?
- What are the components of a Workload description?
- Distinguish between relational OLAP (ROLAP) and multi-dimensional OLAP (MOLAP)

DBMS: Focuses on structured data with entities and relationships

IR System: Focuses on searching through unstructured data with keys

Support is the frequency with which an item set appears in the transactions in the data set. The formula to calculate it is I/T where I is the number of transactions containing the item set and T is the total number of transactions. For an association rule, the item set in question is that which contains the items on the left and right side of the rule.

Confidence is the probability that an item set contains the item on the right side of an association rule given that it contains the item on the left side. The formula to calculate it is $P(A \text{ and } B) / P(A)$ where the association rule is $A \rightarrow B$. This can also be calculated as $\text{support}(\{A,B\})/\text{support}(\{A\})$.

Yes, we need training data for both classification and clustering because both of these frameworks require data to learn either the decision boundary for classification or the cluster centroids for clustering.

Unsure about workload description

OLAP (Online Analytical Processing) queries can involve rollup and drilldown, pivoting (aggregation along selected dimensions), slicing and dicing (equality and range constraints)

OLAP initially caused problems for relational databases because it requires data to be in multidimensional form (like an excel spreadsheet)

MOLAP (multidimensional OLAP) uses an entirely different system than relational databases, with the data in multidimensional form. Alternatively, one can use ROLAP, where you use the existing relational database system and a translator on top. Most MOLAP companies no longer exist.

Q2. a. Give an example where you would want to denormalize your schema.

1. Give an example of an application where you would use (i) Clustering (ii) Classification
2. Describe the k-means algorithm briefly.
3. What are the weaknesses of k-means?

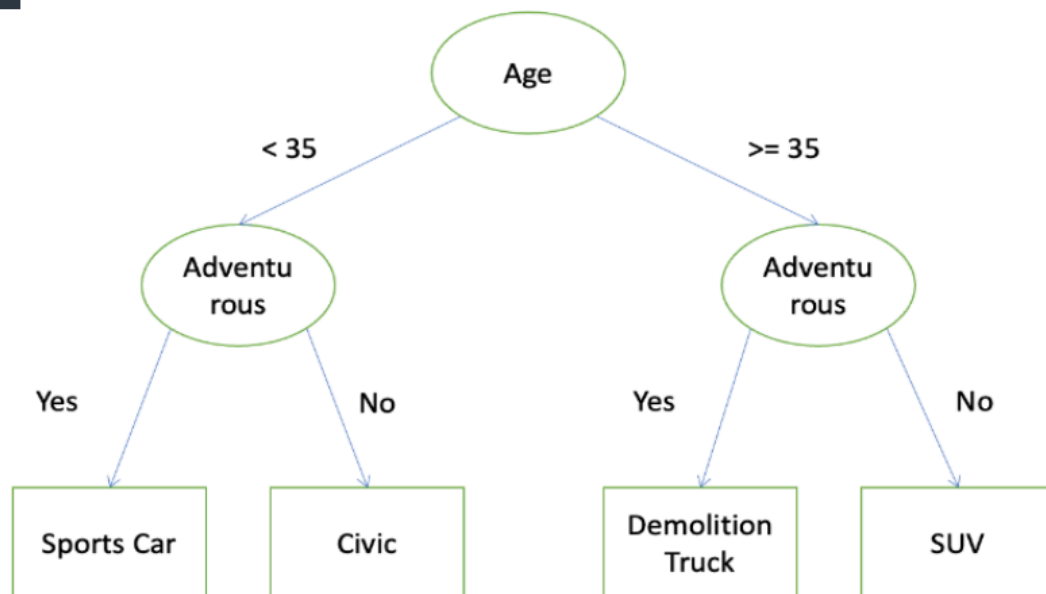
1. You may use clustering to separate 5 different classes of objects based on a set of features. You may use classification to predict whether a transaction is legitimate or fraudulent.

2. The k-means algorithm initializes k centroids at random locations, determines which centroid is closest in terms of euclidean distances for each point and assigns each point to the closest centroid, moves the centroid to the point that minimizes the sum of euclidean distances to the assigned points, then reassigning the points and repeating the process.

3. The weaknesses of k means are that it will not generate good predictions if the true number of clusters differs from the chosen value of k and the clusters will not be accurate if the true clusters are not convex.

3

Essay 1 point



- (a) Age = 10, Nationality = Japanese, Adventurous = No
- (b) Nationality = French, Location = USA, Adventurous = No
- (c) Gender = Female, Nationality = Dutch, Adventurous = Yes, Age = 35
- (d) How would you write this tree in the form of rules?

(e) If you were constructing this tree in a top-down manner, how will you construct the tree?

(f) Describe an approach which can be used to avoid overfitting.

a. Civic

b. ?

c. Demolition Truck

d. if(Adventurous == yes AND Age < 35) Sports Car

if(Adventurous == no AND Age < 35) Civic

if(Adventurous == yes AND Age >= 35) Demolition Truck

if(Adventurous == no AND Age >= 35) SUV

e. I would iteratively split the tree at each node based on which feature split maximizes information gain.

f. To avoid overfitting, you can limit the maximum depth of the tree.

Q4

Consider the table below which represents the document vectors for documents A through I.

<u>DOC</u>	nova	galaxy	heat	hollywood	film	role	diet	legs
A		5	3				3	
B	5	10						
C				11	8	7		4
D				7	7	5		1
E							10	
F					3		9	2
G	2		6					
H		6	8	2	4			2
I			5			1		

- Compute the cosine-similarity between documents C and I.
- Given a query containing the terms {nova, film}, which documents are relevant to the query?
- Rank order the results of (ii) by relevance to the query. (8 pts)

a. cosine similarity = $(C \cdot I) / (||C|| ||I||) = 5 / (15.81 * 5.10) = 0.062$

b. b,c,d,f,g,h

c. $\text{idf}(\text{nova}) = \log(9/2) = 0.653$

$\text{idf}(\text{film}) = \log(9/4) = 0.352$

query $\text{tfidf}(\text{nova}) = 0.653$

query $\text{tfidf}(\text{film}) = 0.352$

doc b $\text{tfidf}(\text{nova}) = 3.265$

doc b $\text{tfidf}(\text{film}) = 0$

doc b relevance = $3.265 * 0.653 = 2.132$

doc c $\text{tfidf}(\text{nova}) = 0$

doc c $\text{tfidf}(\text{film}) = 2.816$

doc c relevance = $2.816 * 0.352 = 0.991$

doc d $\text{tfidf}(\text{nova}) = 0$

doc d $\text{tfidf}(\text{film}) = 2.464$

doc d relevance = $2.464 * 0.352 = 0.867$

doc f $\text{tfidf}(\text{nova}) = 0$

doc f $\text{tfidf}(\text{film}) = 1.056$

doc f relevance = $1.056 * 0.352 = 0.372$

doc g $\text{tfidf}(\text{nova}) = 1.306$

doc g $\text{tfidf}(\text{film}) = 0$

doc g relevance = $1.306 * 0.653 = 0.853$

doc h tfidf(nova) = 0

doc h tfidf(film) = 1.408

doc h relevance = $1.408 * 0.352 = 0.496$

ranked order: b, c, d, g, h, f

Q5

A grocery store manager is interested in finding interesting patterns between

items A, B, C and D in the grocery store. She gives you the transaction matrix below:

Trans.	A	B	C	D
101	1	0	1	0
102	1	1	0	1
103	0	0	1	1
104	1	0	1	1
105	0	1	1	0
106	0	1	0	0
107	1	1	1	1
108	0	1	1	1
109	1	0	1	0
110	1	0	1	1
111	0	1	1	0
112	0	0	1	1

You could help the manager by answering the following questions:

- List all *frequent itemsets*, and their respective support values, which have at least 20% support.
- Suppose a customer buys item D. What is the probability that he will also buy item C? Show your calculations.
- Suppose a customer buys item A. Which other item is she most likely to buy? Why?

Itemsets with at least 20% support: {A}, {B}, {C}, {D}, {A,C}, {A,D}, {B,C}, {B,D}, {C,D}, {A,C,D}

$$P(C|D) = 6/7$$

$$P(B|A) = 2/6$$

$$P(C|A) = 5/6$$

$$P(D|A) = 4/6$$

The customer is most likely to buy item C given that she bought item A because the probability of buying item C is the largest of all items given the purchase of item A